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Information-Based Optimal Subdata Selection for Big Data Linear Regression

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ABSTRACT

Extraordinary amounts of data are being produced in many branches of science. Proven statistical methods are no longer applicable with extraordinary large datasets due to computational limitations. A critical step in big data analysis is data reduction. Existing investigations in the context of linear regression focus on subsampling-based methods. However, not only is this approach prone to sampling errors, it also leads to a covariance matrix of the estimators that is typically bounded from below by a term that is of the order of the inverse of the subdata size. We propose a novel approach, termed information-based optimal subdata selection (IBOSS). Compared to leading existing subdata methods, the IBOSS approach has the following advantages: (i) it is significantly faster; (ii) it is suitable for distributed parallel computing; (iii) the variances of the slope parameter estimators converge to 0 as the full data size increases even if the subdata size is fixed, that is, the convergence rate depends on the full data size; (iv) data analysis for IBOSS subdata is straightforward and the sampling distribution of an IBOSS estimator is easy to assess. Theoretical results and extensive simulations demonstrate that the IBOSS approach is superior to subsampling-based methods, sometimes by orders of magnitude. The advantages of the new approach are also illustrated through analysis of real data. Supplementary materials for this article are available online.

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1. Introduction

Technological advances have enabled an exponential growth in data collection and the size of datasets. Although computational resources have also been growing rapidly, this pales in comparison to the astonishing growth in data volume. This presents the challenge of drawing useful information and converting data into knowledge with available computational resources. We meet this challenge in the context of linear regression by identifying informative subdata, which can be fully analyzed.

For linear regression with an $n \times 1$ response vector and $n \times p$ covariate matrix, the full data volume is $n(p+1)$. In the setting of "big data," this prevents computation of parameter estimates in a traditional way due to insufficient computational resources. The scenario with $p \gg n$ is usually referred to as high-dimensional data. Multiple methods for analyzing high-dimensional data have been proposed and studied, such as LASSO (Tibshirani 1996; Meinshausen et al. 2009), Dantzig selector (Candes and Tao 2007), and sure independence screening (Fan and Lv 2008), among others. We focus on the scenario with $n \gg p$ for an extremely large n. This is an important problem that arises in practice. For example, for the chemical sensors data in Section 5.2, $n = 4,208,261$ and $p = 15$. As another example, the airline on-time dataset from the 2009 ASA Data Expo contains $n = 123,534,969$ observations on 29 variables about flight arrival and departure information for all commercial flights within the USA, from October 1987 to April 2008. Clearly, not everyone has the computing resources to fully analyze the whole data in the aforementioned examples, and thus data reduction is a crucial step to extract useful information from the data. For the case of $n \gg p$, the required computing time for ordinary least squares (OLS) is $O(np^2)$. This time complexity is too long for big data, and may even be beyond the computational capacity of available computing facilities. To address this computational limitation, data reduction is important. Existing investigations in this direction focus on taking random subsamples from the full data. Interesting studies include Drineas, Mahoney, and Muthukrishnan (2006); Drineas et al. (2011); Ma and Sun (2015); Ma, Mahoney, and Yu (2014); Ma et al. (2015), among others. Parameter estimates are obtained based on a small random subsample of the full data. Normalized statistical leverage scores are often used for nonuniform subsampling probabilities, and the method is known as *algorithmic leveraging* (Ma, Mahoney, and Yu 2014).

With exact statistical leverage scores, the computing time of this method is $O(np^2)$, just as for OLS on the full data. Drineas et al. (2012) developed a randomized algorithm to approximate leverage scores; it is $O(np \log n/\epsilon^2)$, and $o(np^2)$ if $\log n = o(p)$, where $\epsilon \in (0, 0.5]$. Thus the computing time for the algorithmic leveraging method is at least $O(np \log n/\epsilon^2)$. A subsampling-based method also induces sampling error and the information in the resultant subdata is typically proportional to its size. We find that (details to be shown in Section 2), for a subsampling-based estimator, the covariance matrix is bounded from below by a term that is typically of order $1/k$, where k is the subsample

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size. This order is only a function of k, so that the variance does not go to 0 with increasing full data size n.

In this article, we propose an alternative subdata selection approach, which we call information-based optimal subdata selection (IBOSS) method from big data. The basic idea is to select the most informative data points deterministically so that subdata of a small size preserves most of the information contained in the full data. It is akin to the basic motivation of optimal experimental design (Kiefer 1959), which aims at obtaining the maximum information with a fixed budget. Traditionally, optimal design is not a data analysis tool, but focuses on data collection. The idea of "maximizing" an information matrix, however, can be borrowed to establish a framework to identify the most informative subdata from the full data for estimating unknown parameters. Using this framework, we will also gain more insight into the popular subsampling-based methods.

As we will show, the IBOSS approach has the following advantages compared to existing methods: (1) the computing time for the IBOSS algorithm is $O(np)$, which is significantly faster than existing methods; (2) the IBOSS algorithm is very suitable for distributed parallel computing platforms; it identifies informative data points by examining each covariate individually; (3) the IBOSS method does not induce sampling error and the variance of estimators can go to 0 as the full data size n becomes large even if the subdata size k is fixed; and (4) in terms of distributional properties, IBOSS estimators inherit properties of estimators based on the full data.

The remainder of the article is organized as follows. In Section 2, we present the IBOSS framework and use it to analyze the popular subsampling-based methods. A lower bound for covariance matrices of subsampling-based estimators will be given. In Section 3, we characterize IBOSS subdata under the D-optimality criterion and use it to develop a computationally efficient algorithm. In Section 4, we evaluate the IBOSS algorithm by deriving its asymptotic properties. In Section 5, the performance of the IBOSS method is examined through extensive simulations and two real data applications. We offer concluding remarks in Section 6 and show all technical details in the online appendix. Additional numerical results are provided in the supplementary material.

2. The Framework

Let $(\mathbf{z}_1, y_1), \ldots, (\mathbf{z}_n, y_n)$ denote the full data, and assume the linear regression model:

$$y_i = \beta_0 + \mathbf{z}_i^{\mathrm{T}}\boldsymbol{\beta}_1 + \varepsilon_i = \beta_0 + \sum_{j=1}^{p} z_{ij}\beta_j + \varepsilon_i, \qquad i = 1, \ldots, n, \tag{1}$$

where β_0 is the scalar intercept parameter, $\boldsymbol{\beta}_1 = (\beta_1, \beta_2, \ldots, \beta_p)^{\mathrm{T}}$ is a p-dimensional vector of unknown slope parameters, $\mathbf{z}_i = (z_{i1}, \ldots, z_{ip})^{\mathrm{T}}$ is a covariate vector, y_i is a response, and ε_i is an error term. We write $\mathbf{x}_i = (1, \mathbf{z}_i^{\mathrm{T}})^{\mathrm{T}}$, $\boldsymbol{\beta} = (\beta_0, \boldsymbol{\beta}_1^{\mathrm{T}})^{\mathrm{T}}$, $\mathbf{Z} = (\mathbf{z}_1, \ldots, \mathbf{z}_n)^{\mathrm{T}}$, $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_n)^{\mathrm{T}}$, $\mathbf{y} = (y_1, \ldots, y_n)^{\mathrm{T}}$, assume that the y_i's are uncorrelated given the covariate matrix $\mathbf{Z}$, and that the error terms ε_i's satisfy $\mathrm{E}(\varepsilon_i) = 0$ and $\mathrm{V}(\varepsilon_i) = \sigma^2$.

The intercept parameter is often not of interest and could be eliminated by centralizing the full data. However, this does not work for streaming data, is not practical if the focus is on building a predictive model, and requires a computing time of $O(np)$.

When using the full data and model (1), the least-square estimator of $\boldsymbol{\beta}$, which is also its best linear unbiased estimator (BLUE), is

$$\hat{\boldsymbol{\beta}}_f = \left(\sum_{i=1}^{n} \mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}\right)^{-1} \sum_{i=1}^{n} \mathbf{x}_i y_i.$$

The covariance matrix of this unbiased estimator is equal to the inverse of

$$\mathbf{M}_f = \frac{1}{\sigma^2}\sum_{i=1}^{n} \mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}.$$

This is the observed Fisher information matrix for $\boldsymbol{\beta}$ from the full data if the ε_i's are normally distributed. While we do not require the normality assumption, for simplicity we will still call $\mathbf{M}_f$ the information matrix.

Let $(\mathbf{z}_1^*, y_1^*), \ldots, (\mathbf{z}_k^*, y_k^*)$ be subdata of size k selected deterministically from the full data, in which the rule to determine whether a data point is included or not depends on $\mathbf{Z}$ only. Then the subdata also follow the linear regression model

$$y_i^* = \beta_0 + \mathbf{z}_i^{*\mathrm{T}}\boldsymbol{\beta}_1 + \varepsilon_i^*, \qquad i = 1, \ldots, k, \tag{2}$$

with the same assumptions and unknown parameters as for model (1). The LS estimator

$$\hat{\boldsymbol{\beta}}_s = \left(\sum_{i=1}^{k} \mathbf{x}_i^*\mathbf{x}_i^{*\mathrm{T}}\right)^{-1} \sum_{i=1}^{k} \mathbf{x}_i^* y_i^*,$$

is the BLUE of $\boldsymbol{\beta}$ for model (2) based on the subdata, where $\mathbf{x}_i^* = (1, \mathbf{z}_i^{*\mathrm{T}})^{\mathrm{T}}$. The observed information matrix for $\boldsymbol{\beta}$ based on the subdata is

$$\mathbf{M}_s = \frac{1}{\sigma^2}\sum_{i=1}^{k} \mathbf{x}_i^*\mathbf{x}_i^{*\mathrm{T}},$$

which is the inverse of the covariance matrix of $\hat{\boldsymbol{\beta}}_s$, namely,

$$\mathrm{V}(\hat{\boldsymbol{\beta}}_s|\mathbf{Z}) = \mathbf{M}_s^{-1} = \sigma^2\left(\sum_{i=1}^{k} \mathbf{x}_i^*\mathbf{x}_i^{*\mathrm{T}}\right)^{-1}. \tag{3}$$

To estimate a linear function of $\boldsymbol{\beta}$ using subdata, plugging in the LS estimator yields the estimator with the minimum variance among all linear unbiased estimators. Since this minimum variance is a function of the covariate values in the subdata, one can judiciously select the subdata to minimize the minimum variance. This is akin to the basic idea behind optimal experimental design (Kiefer 1959). We implement this here by seeking subdata that, in some sense, maximize $\mathbf{M}_s$.

Some additional notation will help to formulate this idea as an optimization problem. Let δ_i be the indicator variable that signifies whether $(\mathbf{z}_i, y_i)$ is included in the subdata, that is,

$\delta_i = 1$ if $(\mathbf{z}_i, y_i)$ is included and $\delta_i = 0$ otherwise. The information matrix with subdata of size k can then be written as

$$\mathbf{M}(\boldsymbol{\delta}) = \frac{1}{\sigma^2}\sum_{i=1}^{n}\delta_i\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}, \tag{4}$$

where $\boldsymbol{\delta} = \{\delta_1, \delta_2, \ldots, \delta_n\}$ such that $\sum_{i=1}^{n}\delta_i = k$. To have an optimal estimator based on a subdata, one can choose a $\boldsymbol{\delta}$ that "maximizes" the information matrix (4). Since $\mathbf{M}(\boldsymbol{\delta})$ is a matrix, in optimal experimental design (Kiefer 1959), this is typically done by maximizing a univariate *optimality criterion function* of the matrix.

Let ψ denote an optimality criterion function. The problem is presented as the following optimization problem given the observed big data:

$$\boldsymbol{\delta}^{\mathrm{opt}} = \arg\max_{\boldsymbol{\delta}} \psi\{\mathbf{M}(\boldsymbol{\delta})\}, \quad \text{subject to} \quad \sum_{i=1}^{n}\delta_i = k. \tag{5}$$

A popular optimality criterion is the D-optimality criterion. It maximizes the determinant of $\mathbf{M}(\boldsymbol{\delta})$, which has the interpretation of minimizing the expected volume of the joint confidence ellipsoid for $\boldsymbol{\beta}$. We will come back to this criterion in greater detail in Section 3.

2.1. Analysis of Existing Subsampling-Based Methods

Based on our information-based subdata selection framework, we can also gain insights into popular random subsampling-based methods. To see this, let $\boldsymbol{\eta}_L$ be the n-dimensional count-vector whose ith entry denotes the number of times that the ith data point is included in a subsample of size k, which is obtained using a random subsampling method with probabilities proportional to π_i, $i = 1, \ldots, n$, such that $\sum_{i=1}^{n}\pi_i = 1$. A subsampling-based estimator has the general form

$$\tilde{\boldsymbol{\beta}}_L = \left(\sum_{i=1}^{n} w_i\eta_{Li}\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}\right)^{-1}\sum_{i=1}^{n} w_i\eta_{Li}\mathbf{x}_i y_i, \tag{6}$$

where the weight w_i is often taken to be $1/\pi_i$. Corresponding to different choices of π_i and w_i, some popular subsampling-based methods (Ma et al. 2015) include: uniform subsampling (UNI) in which $\pi_i = 1/n$ and $w_i = 1$; leverage-based subsampling (LEV) in which $\pi_i = h_{ii}/(p+1)$, $w_i = 1/\pi_i$, and $h_{ii} = \mathbf{x}_i^{\mathrm{T}}(\mathbf{X}\mathbf{X}^{\mathrm{T}})^{-1}\mathbf{x}_i$; shrinked leveraging estimator (SLEV) in which $\pi_i = \alpha h_{ii}/(p+1) + (1-\alpha)/n$, $w_i = 1/\pi_i$, and $\alpha \in [0, 1]$; unweighted leveraging estimator (LEVUNW) in which $\pi_i = h_{ii}/(p+1)$ and $w_i = 1$.

The distribution of $\tilde{\boldsymbol{\beta}}_L$ is complicated, but we can study its performance using the proposed information-based framework. The "information matrix" $\mathbf{M}(\boldsymbol{\eta}_L)$ given $\mathbf{Z}$ is random because $\boldsymbol{\eta}_L$ is random. While the η_{Li}'s are correlated, we will only need to use the marginal distribution of each η_{Li}. If subsampling is with replacement, then each η_{Li} has a binomial distribution $\mathrm{Bin}(k, \pi_i)$; if the subsampling is without replacement, the marginal distribution of each η_{Li} is $\mathrm{Bin}(1, k\pi_i)$ under the condition that $k\pi_i \le 1$. Either way, $\mathrm{E}(\eta_{Li}) = k\pi_i$. Hence, taking expectations with respect to $\boldsymbol{\eta}_L$, the expected observed information matrix for a subsampling-based method is

$$\mathbf{M}_{EL} = \mathrm{E}\{\mathbf{M}(\boldsymbol{\eta}_L)|\mathbf{Z}\} = \frac{1}{\sigma^2}\sum_{i=1}^{n}\mathrm{E}(\eta_{Li})\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}} = \frac{k}{\sigma^2}\sum_{i=1}^{n}\pi_i\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}. \tag{7}$$

Unlike the IBOSS approach, the inverse of $\mathbf{M}_{EL}$ is not the variance covariance matrix of $\mathrm{V}(\tilde{\boldsymbol{\beta}}_L|\mathbf{Z})$. In fact, for subsampling with replacement there is a small probability that $\mathrm{V}(\tilde{\boldsymbol{\beta}}_L|\mathbf{Z})$ is not well defined because $\boldsymbol{\beta}_L$ is not estimable. To solve this issue, we consider only subsamples with full-rank covariate matrices to define the covariance matrix of $\tilde{\boldsymbol{\beta}}_L$. The following theorem states a relationship between $\mathbf{M}_{EL}$ and the covariance matrix of $\tilde{\boldsymbol{\beta}}_L$.

Theorem 1. Suppose that a subsample of size k is taken using a random subsampling procedure with probabilities proportional to π_i, $i = 1, \ldots, n$, such that $\sum_{i=1}^{n}\pi_i = 1$. Consider the set $\Delta = \{\boldsymbol{\eta}_L : \sum_{i=1}^{n}\eta_{Li}\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}$ is nonsingular$\}$, where $\boldsymbol{\eta}_L$ is the n-dimensional vector that counts how often each data point is included. Let $I_\Delta(\boldsymbol{\eta}_L) = 1$ if and only if $\boldsymbol{\eta}_L \in \Delta$. Given $I_\Delta(\boldsymbol{\eta}_L) = 1$, $\tilde{\boldsymbol{\beta}}_L$ is unbiased for $\boldsymbol{\beta}$, and

$$\begin{aligned}\mathrm{V}\{\tilde{\boldsymbol{\beta}}_L|\mathbf{Z}, I_\Delta(\boldsymbol{\eta}_L) = 1\} &\ge P\{I_\Delta(\boldsymbol{\eta}_L) = 1|\mathbf{Z}\}\mathbf{M}_{EL}^{-1} \\ &= \frac{\sigma^2 P\{I_\Delta(\boldsymbol{\eta}_L) = 1|\mathbf{Z}\}}{k}\left\{\sum_{i=1}^{n}\pi_i\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}\right\}^{-1}\end{aligned} \tag{8}$$

in the Loewner ordering.

Remark 1. Theorem 1 is true regardless of the choice for the weights w_i's in $\tilde{\boldsymbol{\beta}}_L$ or whether subsampling is with or without replacement. It provides a lower bound for covariance matrices of subsampling-based estimators, which provides a feasible way to evaluate the best performance of a subsampling-based estimator. Note that when $k \gg p+1$, $P\{I_\Delta(\boldsymbol{\eta}_L) = 1|\mathbf{Z}\}$ is often close to 1, so that the lower bound is close to the inverse of the expected observed information $\mathbf{M}_{EL}$.

Remark 2. Existing investigations on the random subsampling approach focus on sampling with replacement where the subsample is independent given the full data. For subsampling without replacement with fixed sample size, the subsample is no longer independent given the full data and the properties of the resultant estimator are more complicated. As a result, to the best of our knowledge, subsampling without replacement for a fixed subsample size from big data has never been investigated. Our IBOSS framework, however, is applicable here to assess the performance of an estimator based on subsampling without replacement.

Applying Theorem 1 to the popular sampling-based methods, we obtain the following results. For the UNI method,

$$\mathrm{V}\left\{\tilde{\boldsymbol{\beta}}_L^{\mathrm{UNI}}\middle|\mathbf{Z}, I_\Delta(\boldsymbol{\eta}_L) = 1\right\} \ge \frac{\sigma^2 P\{I_\Delta(\boldsymbol{\eta}_L) = 1|\mathbf{Z}\}}{k}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}\right)^{-1}; \tag{9}$$

for the LEV (or LEVUNW) method,

$$\mathrm{V}\left\{\tilde{\boldsymbol{\beta}}_L^{\mathrm{LEV}}\middle|\mathbf{Z}, I_\Delta(\boldsymbol{\eta}_L)=1\right\} \geq \frac{(p+1)\sigma^2 P\{I_\Delta(\boldsymbol{\eta}_L)=1|\mathbf{Z}\}}{k} \times \left\{\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}\right\}^{-1}; \tag{10}$$

for the SLEV method,

$$\mathrm{V}\left\{\tilde{\boldsymbol{\beta}}_L^{\mathrm{SLEV}}\middle|\mathbf{Z}, I_\Delta(\boldsymbol{\eta}_L)=1\right\} \geq \frac{\sigma^2 P\{I_\Delta(\boldsymbol{\eta}_L)=1|\mathbf{Z}\}}{k}\left\{\frac{\alpha}{p+1}\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}} + \frac{1-\alpha}{n}\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}\right\}^{-1}. \tag{11}$$

If $\mathbf{x}_i$, $i=1,\ldots,n$, are generated independently from the same distribution as random vector $\mathbf{x}$ with finite second moment, that is, $\mathrm{E}\|\mathbf{x}\|^2<\infty$, then from the strong law of large numbers,

$$\frac{1}{n}\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^{\mathrm{T}} \to \mathrm{E}(\mathbf{x}\mathbf{x}^{\mathrm{T}}), \tag{12}$$

almost surely as $n\to\infty$. If we further assume that the fourth moment of $\mathbf{x}$ is finite, that is, $\mathrm{E}\|\mathbf{x}\|^4<\infty$, then we have

$$\sum_{i=1}^n \mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}} \to \mathrm{E}\left[\mathbf{x}\mathbf{x}^{\mathrm{T}}\{\mathrm{E}(\mathbf{x}\mathbf{x}^{\mathrm{T}})\}^{-1}\mathbf{x}\mathbf{x}^{\mathrm{T}}\right] \tag{13}$$

almost surely as $n\to\infty$. Let $P_\eta=\liminf_{n\to\infty}P\{I_\Delta(\boldsymbol{\eta}_L)=1|\mathbf{Z}\}$. Note that $P_\eta=1$ under some mild condition, for example, the covariate distribution is continuous. From (9) to (13), we have

$$\mathrm{V}\left\{\tilde{\boldsymbol{\beta}}_L^{\mathrm{UNI}}\middle|\mathbf{Z}, I_\Delta(\boldsymbol{\eta}_L)=1\right\} \geq \frac{\sigma^2 P_\eta}{k}\left\{\mathrm{E}(\mathbf{x}\mathbf{x}^{\mathrm{T}})\right\}^{-1}, \tag{14}$$

$$\mathrm{V}\left\{\tilde{\boldsymbol{\beta}}_L^{\mathrm{LEV}}\middle|\mathbf{Z}, I_\Delta(\boldsymbol{\eta}_L)=1\right\} \geq \frac{(p+1)\sigma^2 P_\eta}{k} \times \left(\mathrm{E}[\mathbf{x}\mathbf{x}^{\mathrm{T}}\{\mathrm{E}(\mathbf{x}\mathbf{x}^{\mathrm{T}})\}^{-1}\mathbf{x}\mathbf{x}^{\mathrm{T}}]\right)^{-1}, \tag{15}$$

$$\mathrm{V}\left\{\tilde{\boldsymbol{\beta}}_L^{\mathrm{SLEV}}\middle|\mathbf{Z}, I_\Delta(\boldsymbol{\eta}_L)=1\right\} \geq \frac{\sigma^2 P_\eta}{k}\left(\frac{\alpha}{p+1}\mathrm{E}[\mathbf{x}\mathbf{x}^{\mathrm{T}}\{\mathrm{E}(\mathbf{x}\mathbf{x}^{\mathrm{T}})\}^{-1}\mathbf{x}\mathbf{x}^{\mathrm{T}}] + (1-\alpha)\mathrm{E}(\mathbf{x}\mathbf{x}^{\mathrm{T}})\right)^{-1}, \tag{16}$$

almost surely as $n\to\infty$.

From (14), (15), and (16), one sees that covariance matrices of these commonly used subsampling-based estimators are bounded from below in the Loewner ordering by finite quantities that are at the order of $1/k$. These quantities do not go to 0 as the full data sample size n goes to ∞.

3. The D-Optimality Criterion and An IBOSS Algorithm

In this section, we study the commonly used D-optimality criterion and develop IBOSS algorithms based on theoretical characterizations of IBOSS subdata under this criterion.

In our framework, for given full data of size n, the D-optimality criterion suggests the selection of subdata of size k so that

$$\boldsymbol{\delta}_{\mathrm{D}}^{\mathrm{opt}} = \arg\max_{\boldsymbol{\delta}}\left|\sum_{i=1}^n \delta_i\mathbf{x}_i\mathbf{x}_i^{\mathrm{T}}\right|, \quad \sum_{i=1}^n \delta_i = k.$$

Obtaining an exact solution is computationally far too expensive. In working toward an approximate solution, we first derive an upper bound for $|\mathbf{M}(\boldsymbol{\delta})|$ which, while only attainable for very special cases, will guide our later algorithm.

Theorem 2 (D-optimality). For subdata of size k represented by $\boldsymbol{\delta}$,

$$|\mathbf{M}(\boldsymbol{\delta})| \leq \frac{k^{p+1}}{4^p\sigma^{2(p+1)}}\prod_{j=1}^p (z_{(n)j}-z_{(1)j})^2, \tag{17}$$

where $z_{(n)j}=\max\{z_{1j}, z_{2j},\ldots,z_{nj}\}$ and $z_{(1)j}=\min\{z_{1j}, z_{2j},\ldots,z_{nj}\}$ are the nth and first-order statistics of $z_{1j}, z_{2j},\ldots,z_{nj}$. If the subdata consists of the 2^p points $(a_1,\ldots,a_p)^{\mathrm{T}}$ where $a_j=z_{(n)j}$ or $z_{(1)j}$, $j=1,2,\ldots,p$, each occurring equally often, then equality holds in (17).

Remark 3. Often k is much smaller than 2^p, so that subdata with equality in Theorem 2 will not exist. However, just as for Hadamard's determinant bound (Hadamard 1893), the result suggests to collect subdata with extreme covariate values, both small and large, occurring with the same frequency.

This agrees with the common statistical knowledge that larger variation in covariates is more informative and results in better parameter estimation.

The following algorithm is motivated by the result in Theorem 2.

Algorithm 1 (Algorithm motivated by D-optimality). Suppose that $r=k/(2p)$ is an integer. Using a partition-based selection algorithm, Martínez (2004), perform the following steps:

1. For z_{i1}, $1\leq i\leq n$, include r data points with the r smallest z_{i1} values and r data points with the r largest z_{i1} values;
2. For $j=2,\ldots,p$, exclude data points that were previously selected, and from the remainder select r data points with the smallest z_{ij} values and r data points with the largest z_{ij} values.
3. Return $\hat{\boldsymbol{\beta}}^{\mathrm{D}}=\{(\mathbf{X}_{\mathrm{D}}^*)^{\mathrm{T}}\mathbf{X}_{\mathrm{D}}^*\}^{-1}(\mathbf{X}_{\mathrm{D}}^*)^{\mathrm{T}}\mathbf{y}_{\mathrm{D}}^*$ and the estimated covariance matrix $\hat{\sigma}_{\mathrm{D}}^2\{(\mathbf{X}_{\mathrm{D}}^*)^{\mathrm{T}}\mathbf{X}_{\mathrm{D}}^*\}^{-1}$, where $\mathbf{X}_{\mathrm{D}}^*=(\mathbf{1},\mathbf{Z}_{\mathrm{D}}^*)$, $\mathbf{Z}_{\mathrm{D}}^*$ is the covariate matrix of the subdata selected in the previous steps, $\mathbf{y}_{\mathrm{D}}^*$ is the response vector of the subdata, and $\hat{\sigma}_{\mathrm{D}}^2=\left\|\mathbf{y}_{\mathrm{D}}^*-\mathbf{X}_{\mathrm{D}}^*\hat{\boldsymbol{\beta}}^{\mathrm{D}}\right\|^2/(k-p-1)$.

Remark 4. For each covariate, a partition-based selection algorithm has an average time complexity of $O(n)$ to find the rth largest or smallest value (Musser 1997; Martínez 2004).

Thus the time to obtain the subdata is $O(np)$. Using the subdata, the computing time for $\hat{\boldsymbol{\beta}}^{\mathrm{D}}$ and $\hat{\sigma}_{\mathrm{D}}^2$ is $O(kp^2+p^3)$ and $O(kp)$, respectively. Thus, the time complexity of Algorithm 1 is $O(np+kp^2+p^3+kp)=O(np+kp^2)$. For the scenario that

$n > kp$, this reduces to $O(np)$. This algorithm is faster than algorithmic leveraging, which has a computing time of $O(np \log n)$ (Drineas et al. 2012).

Remark 5. Algorithm 1 gives the covariance matrix of the resultant estimator, which is very crucial for statistical inference. This is the exact covariance matrix of $\hat{\boldsymbol{\beta}}^{\mathrm{D}}$ if the variance of the error term, σ^2, is known. With an additional assumption of normality of ε_i, $\hat{\boldsymbol{\beta}}^{\mathrm{D}}$ has an exact normal distribution.

Remark 6. Algorithm 1 is naturally suited for distributed storage and processing facilities for parallel computing. One can simultaneously process each covariate and find the indexes of its extreme values. These indexes can then be combined to obtain the subdata. While this approach may result in a subdata size that is smaller than k if there is duplication of indexes, the resultant estimator will still have the same convergence rate.

Remark 7. Algorithm 1 selects subdata according to extreme values of each covariate, which may include outliers. However, the selection rule is ancillary, and the resultant subdata follow the same underlying regression model as the full data. We can thus use outlier diagnostic methods to identify outliers in the subdata. If there are outliers in the full data, it is very likely that these data points will be identified as outliers in the subdata. On the other hand, if there are data points that are far from others but still follow the underlying model, then these data points actually contain more information about the model and should be used for parameter estimation.

Remark 8. The restriction that the subdata sample size k is chosen to make $r = k/(2p)$ an integer is mostly for convenience. In the case that $r = k/(2p)$ is not an integer, one can either adjust k by using the floor $\lfloor r \rfloor$ or the ceiling $\lceil r \rceil$, or use a combination of $\lfloor r \rfloor$ and $\lceil r \rceil$ to keep the subdata sample size as k.

The following theorem gives some insight on the quality of using Algorithm 1 to approximate the upper bound of $|\mathbf{M}(\boldsymbol{\delta})|$ in Theorem 2.

Theorem 3. Let $\mathbf{Z}_{\mathrm{D}}^*$ be the covariate matrix for the subdata of size $k = 2pr$ selected using Algorithm 1 and $\mathbf{X}_{\mathrm{D}}^* = (\mathbf{1}, \mathbf{Z}_{\mathrm{D}}^*)$. The determinant $|(\mathbf{X}_{\mathrm{D}}^*)^{\mathrm{T}}\mathbf{X}_{\mathrm{D}}^*|$ satisfies

$$\frac{|(\mathbf{X}_{\mathrm{D}}^*)^{\mathrm{T}}\mathbf{X}_{\mathrm{D}}^*|}{\frac{k^{p+1}}{4^p}\prod_{j=1}^p (z_{(n)j} - z_{(1)j})^2} \geq \frac{\lambda_{\min}^p(\mathbf{R})}{p^p}\prod_{j=1}^p \left(\frac{z_{(n-r+1)j} - z_{(r)j}}{z_{(n)j} - z_{(1)j}}\right)^2, \tag{18}$$

where $\lambda_{\min}(\mathbf{R})$ is the smallest eigenvalue of $\mathbf{R}$, the sample correlation matrix of $\mathbf{Z}_{\mathrm{D}}^*$.

From this theorem, it is seen that although Algorithm 1 may not achieve the unachievable upper bound in Theorem 2, it may achieve the same order. For example, if p is fixed and $\liminf_{n\to\infty} \lambda_{\min}(\mathbf{R}) > 0$, then under reasonable assumptions, the lower bound in Equation (18) will not converge to 0 as $n \to \infty$. This means that $|(\mathbf{X}_{\mathrm{D}}^*)^{\mathrm{T}}\mathbf{X}_{\mathrm{D}}^*|$ is of the same order as the upper bound for $|\mathbf{M}(\boldsymbol{\delta})|$ in Theorem 2, even though the latter is typically not attainable.

4. Properties of Parameter Estimator

In this section, we investigate the theoretical properties of the D-optimality motivated IBOSS algorithm, and provide both finite sample assessment and asymptotic results. These results are provided to evaluate the performance of the proposed method and show more insights about the IBOSS approach. The application of the D-optimality motivated IBOSS algorithm does not depend on asymptotic properties of the approach.

Since $\hat{\boldsymbol{\beta}}^{\mathrm{D}}$ is unbiased for $\boldsymbol{\beta}$, we focus on its variance. The next theorem gives bounds on variances of estimators of the intercept and slope parameters from the D-optimality-motivated algorithm.

Theorem 4. If $\lambda_{\min}(\mathbf{R}) > 0$, then, the following results hold for the estimator, $\hat{\boldsymbol{\beta}}^{\mathrm{D}}$, obtained from Algorithm 1:

$$\mathrm{V}(\hat{\beta}_0^{\mathrm{D}}|\mathbf{Z}) \geq \frac{\sigma^2}{k}, \tag{19}$$

$$\frac{4\sigma^2}{k\lambda_{\max}(\mathbf{R})(z_{(n)j} - z_{(1)j})^2} \leq \mathrm{V}(\hat{\beta}_j^{\mathrm{D}}|\mathbf{Z}) \leq \frac{4p\sigma^2}{k\lambda_{\min}(\mathbf{R})(z_{(n-r+1)j} - z_{(r)j})^2}, \quad j = 1, \ldots, p. \tag{20}$$

Theorem 4 describes finite sample properties of the proposed estimator and does not require any quantity to go to ∞. It shows that the variance of the intercept estimator is bounded from below by a term proportional to the inverse subdata size. This is similar to the results for existing subsampling methods. However, for the slope estimator, the variance is bounded from above by a term that is proportional to $\frac{p}{k(z_{(n-r+1)j} - z_{(r)j})^2}$, which may converge to 0 as n increases even when the subdata size k is fixed. We present this asymptotic result in the following theorem.

Theorem 5. Assume that covariate distributions are in the domain of attraction of the generalized extreme value distribution, and $\liminf_{n\to\infty} \lambda_{\min}(\mathbf{R}) > 0$. For large enough n, the following results hold for the estimator, $\hat{\boldsymbol{\beta}}^{\mathrm{D}}$, obtained from Algorithm 1:

$$\mathrm{V}(\hat{\beta}_j^{\mathrm{D}}|\mathbf{Z}) = O_P\left\{\frac{p}{k(z_{(n-r+1)j} - z_{(r)j})^2}\right\}, \quad j = 1, \ldots, p. \tag{21}$$

Furthermore,

$$\mathrm{V}(\hat{\beta}_j^{\mathrm{D}}|\mathbf{Z}) \asymp_P \frac{p}{k(z_{(n)j} - z_{(1)j})^2}, \quad j = 1, \ldots, p, \tag{22}$$

if one of the following conditions holds: (1) r is fixed; (2) the support of F_j is bounded, $r \to \infty$, and $r/n \to 0$, where F_j is the marginal distribution function of the jth component of $\mathbf{z}$; (3) the upper endpoint for the support of F_j is ∞ and the lower endpoint for the support of F_j is finite, and $r \to \infty$ slow enough such that

$$\frac{r}{n[1 - F_j\{(1-\epsilon)F_j^{-1}(1 - n^{-1})\}]} \to 0, \tag{23}$$

for all $\epsilon > 0$; (4) the upper endpoint for the support of F_j is finite and the lower endpoint for the support of F_j is $-\infty$, and $r \to \infty$ slow enough such that

$$\frac{r}{nF_j\{(1-\epsilon)F_j^{-1}(n^{-1})\}} \to 0, \tag{24}$$

for all $\epsilon > 0$; (5) the upper endpoint and the lower endpoint for the support of F_j are ∞ and $-\infty$, respectively, and (23) and (24) hold.

Equation (21) gives a general result on the variance of a slope estimator. It holds for any values of n, r, and p, so that it can also be used to obtain asymptotic results when one or more of n, r, and p go to infinity. The expression shows that if $p/(z_{(n-r+1)j} - z_{(r)j})^2 = o_P(1)$, then the convergence of the variance would be faster than $1/k$, the typical convergence rate for a subsampling method (Ma et al. 2015; Wang, Zhu, and Ma 2017). Note that the results are derived from the upper bound in (20), and thus the real convergence of the variance can be faster than $\frac{p}{k(z_{(n-r+1)j}-z_{(r)j})^2}$.

For the condition in (23), it can be satisfied by many commonly seen distributions, such as exponential distribution, double exponential distribution, lognormal distribution, normal distribution, and gamma distribution (Hall 1979). For different distributions, the required rate at which $r \to \infty$ is different. For example, if F_j is a normal distribution function, then (23) holds if and only if $\log r/\log\log n \to 0$; if F_j is an exponential distribution function, then (23) holds if and only if $\log r/\log n \to 0$. The condition in (24) is the same as that in (23) if one takes $\mathbf{z} = -\mathbf{z}$.

For the result in (22), it can be shown from the proof that

$$\mathrm{V}(\hat{\beta}_j^{\mathrm{D}}|\mathbf{Z}) \asymp_P \frac{p}{k\left\{F_j^{-1}(1-n^{-1}) - F_j^{-1}(n^{-1})\right\}^2}, \quad j = 1, \ldots, p.$$

What we find more interesting is the fact that, when k is fixed, from Theorems 2.8.1 and 2.8.2 in Galambos (1987), $z_{(n-r+1)j} - z_{(r)j}$ goes to infinity with the same rate as that of $z_{(n)j} - z_{(1)j}$. Thus, the order of the variance of a slope estimator is the inverse of the squared full data sample range for the corresponding covariate. If the sample range goes to ∞ as $n \to \infty$, then the variance converges to 0 even when the subdata size k is fixed. This suggests that subdata may preserve information at a scale related to the full data size. We will return to this for specific cases with more details. In the remainder of this section, we focus on the case that both p and k are fixed.

That the variance $\mathrm{V}(\hat{\beta}_0^{\mathrm{D}}|\mathbf{Z})$ does not go to 0 for a fixed subdata size k is not a concern if inference for the slope parameters is of primary interest, as is often the case. However, if the focus is on building a predictive model, the intercept needs to be estimated more precisely. This can be done by using the full data means, $\bar{y}$ and $\bar{\mathbf{z}}$, say. After obtaining the slope estimator $\hat{\boldsymbol{\beta}}_1^{\mathrm{D}}$, compute the following adjusted estimator of the intercept:

$$\hat{\beta}_0^{Da} = \bar{y} - \bar{\mathbf{z}}^{\mathrm{T}}\hat{\boldsymbol{\beta}}_1^{\mathrm{D}}. \tag{25}$$

The estimator $\hat{\beta}_0^{Da}$ has a convergence rate similar to that of the slope parameter estimators, because $\hat{\beta}_0^{Da} - \beta_0 = (\hat{\beta}_0^{\mathrm{full}} - \beta_0) + \bar{\mathbf{z}}^{\mathrm{T}}(\hat{\boldsymbol{\beta}}_1^{\mathrm{full}} - \boldsymbol{\beta}_1) - \bar{\mathbf{z}}^{\mathrm{T}}(\hat{\boldsymbol{\beta}}_1^{\mathrm{D}} - \boldsymbol{\beta}_1)$ and the last term is the dominating term if $\mathrm{E}(\mathbf{z}) \neq \mathbf{0}$. The rate may be faster than that of the slope parameter estimators if $\mathrm{E}(\mathbf{z}) = \mathbf{0}$. The additional computing time for this approach is $O(np)$, but the estimation efficiency for β_0 will be substantially improved. We will demonstrate this numerically in Section 5.

Whereas Theorem 5 provides a general result for the variance of individual parameter estimators, more can be said for special cases. The next theorem studies the structure of the covariance matrix for estimators based on Algorithm 1 under various assumptions.

Theorem 6. Let $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_p)^{\mathrm{T}}$ and let $\boldsymbol{\Sigma} = \boldsymbol{\Phi}\boldsymbol{\rho}\boldsymbol{\Phi}$ be a full rank covariance matrix, where $\boldsymbol{\Phi} = \mathrm{diag}(\sigma_1, \ldots, \sigma_p)$ is a diagonal matrix of standard deviations and $\boldsymbol{\rho}$ is a correlation matrix. Assume that $\mathbf{z}_i$'s, $i = 1, \ldots, n$, are iid with a distribution specified below. The following results hold for $\hat{\boldsymbol{\beta}}^{\mathrm{D}}$, the estimator from Algorithm 1.

(i) For multivariate normal covariates, that is, $\mathbf{z}_i \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

$$\mathrm{V}(\mathbf{A}_n\hat{\boldsymbol{\beta}}^{\mathrm{D}}|\mathbf{Z}) = \frac{\sigma^2}{2k}\begin{bmatrix} 2 & \mathbf{0} \\ \mathbf{0} & p(\boldsymbol{\Phi}\boldsymbol{\rho}^2\boldsymbol{\Phi})^{-1} \end{bmatrix} + O_P\left(\frac{1}{\sqrt{\log n}}\right), \tag{26}$$

where $\mathbf{A}_n = \mathrm{diag}\big(1, \sqrt{\log n}, \ldots, \sqrt{\log n}\big)$.

(ii) For multivariate lognormal covariates, that is, $\mathbf{z}_i \sim \mathrm{LN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

$$\mathrm{V}(\mathbf{A}_n\hat{\boldsymbol{\beta}}^{\mathrm{D}}|\mathbf{Z}) = \frac{2\sigma^2}{k}\begin{bmatrix} 1 & -\mathbf{u}^{\mathrm{T}} \\ -\mathbf{u} & p\boldsymbol{\Lambda} + \mathbf{u}\mathbf{u}^{\mathrm{T}}, \end{bmatrix} + o_P(1), \tag{27}$$

where $\mathbf{A}_n = \mathrm{diag}\ \{1, \exp\ (\sigma_1\ \sqrt{2\log n}), \ldots, \exp\ (\sigma_p\ \sqrt{2\log n})\}$, $\mathbf{u} = (e^{-\mu_1}, \ldots, e^{-\mu_p})^{\mathrm{T}}$, and $\boldsymbol{\Lambda} = \mathrm{diag}\ (e^{-2\mu_1}, \ldots, e^{-2\mu_p})$.

For the distributions in Theorem 6, $\mathrm{V}(\hat{\beta}_0^{\mathrm{D}}|\mathbf{Z})$ is proportional to $1/k$ and never converges to 0 with a fixed k. Based on Theorem 6, $\mathrm{V}(\hat{\boldsymbol{\beta}}_1^{\mathrm{D}}|\mathbf{Z})$ converges to 0 at different rates for different distributions. When $\mathbf{z}$ has a normal distribution, the convergence rate of components of $\mathrm{V}(\hat{\boldsymbol{\beta}}_1^{\mathrm{D}}|\mathbf{Z})$ is $1/\log n$. When $\mathbf{z}$ has a lognormal distribution, the component $\mathrm{V}_{j_1 j_2}(\hat{\boldsymbol{\beta}}_1^{\mathrm{D}}|\mathbf{Z})$ has a convergence rate $\exp\left\{-(\sigma_{j_1} + \sigma_{j_2})\sqrt{2\log n}\right\}$, $j_1, j_2 = 1, \ldots, p$. In comparison, for most popular subsampling-based methods, from the results in Section 2.1, for the normal and lognormal distributions, variances of the slope parameter estimators never converge to 0 because they are bounded from below by terms that are proportional to $1/k$.

For a subsampling-based method, if the covariate distribution is sufficiently heavy-tailed, then some components of the lower bound in Theorem 1 can go to 0. However, even then the convergence rate is much slower than that for the IBOSS approach, which may then produce an estimator with a convergence rate close to that of the full data estimator. For example, Table 1 summarizes the orders of variances for parameter estimators when the only covariate z in a simple linear regression model has a t distribution with degrees of freedom ν. Three approaches are compared: the D-optimality-motivated IBOSS approach (D-OPT), the UNI approach, and the full data approach (FULL).

Table 1. Orders of variances and orders of lower bounds of variances when the covariate has a t_ν distribution. The orders are in probability.

	Covariates are t_ν		
		β_1	
Methods	β_0	$\nu > 2$	$\nu \le 2$
D-OPT	$1/k$	$1/(kn^{2/\nu})$	$1/(kn^{2/\nu})$
UNI	$1/k$	$1/k$	slower than $1/\{kn^{(2/\nu-1+\alpha)}\}$ for any $\alpha > 0$
FULL	$1/n$	$1/n$	slower than $1/\{kn^{(2/\nu+\alpha)}\}$ for any $\alpha > 0$

For β_0, neither subdata approach produces a variance that goes to 0. For β_1, the D-OPT IBOSS approach results in a variance that goes to 0 at a rate of $n^{-2/\nu}$. When $\nu \le 2$, the variance of the estimator based on the full data goes to 0 at a rate that is slower than $n^{-(2/\nu+\alpha)}$ for any $\alpha > 0$, so that the D-OPT IBOSS approach reaches a rate that is very close to that of the full data. For the UNI approach, the lower bound of the variance goes to 0 at a much slower rate. Note that the convergence to 0 does not contradict the conclusion in (14), which assumes that the $\mathbf{x}_i$ are iid with finite second moment.

5. Numerical Experiments

Using simulated and real data, we will now evaluate the performance of the IBOSS method.

5.1. Simulation Studies

Data are generated from the linear model (1) with the true value of $\boldsymbol{\beta}$ being a 51-dimensional vector of unity and $\sigma^2 = 9$. An intercept is included so $p = 50$. Let $\boldsymbol{\Sigma}$ be a covariance matrix with $\Sigma_{ij} = 0.5^{I(i\neq j)}$, for $i, j = 1, \ldots, 50$, where $I()$ is the indicator function. Covariates $\mathbf{z}_i$'s are generated according to the following scenarios.

Case 1. $\mathbf{z}_i$'s have a multivariate normal distribution, that is, $\mathbf{z}_i \sim N(\mathbf{0}, \boldsymbol{\Sigma})$.

Case 2. $\mathbf{z}_i$'s have a multivariate lognormal distribution, that is, $\mathbf{z}_i \sim \mathrm{LN}(\mathbf{0}, \boldsymbol{\Sigma})$.

Case 3. $\mathbf{z}_i$'s have a multivariate t distribution with degrees of freedom $\nu = 2$, that is, $\mathbf{z}_i \sim t_2(\mathbf{0}, \boldsymbol{\Sigma})$.

Case 4. $\mathbf{z}_i$'s have a mixture distribution of five different distributions, $N(\mathbf{1}, \boldsymbol{\Sigma})$, $t_2(\mathbf{1}, \boldsymbol{\Sigma})$, $t_3(\mathbf{1}, \boldsymbol{\Sigma})$, $\mathrm{U}[\mathbf{0}, \mathbf{2}]$, and $\mathrm{LN}(\mathbf{0}, \boldsymbol{\Sigma})$ with equal proportions, where $\mathrm{U}[\mathbf{0}, \mathbf{2}]$ means its components are independent uniform distributions between 0 and 2.

Case 5. $\mathbf{z}_i$'s consist of multivariate normal random variables with interactions and quadratic terms. To be specific, denote $\mathbf{v} = (v_1, \ldots, v_{20})^{\mathrm{T}} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_{20\times 20})$, where $\boldsymbol{\Sigma}_{20\times 20}$ is the 20 by 20 upper diagonal sub-matrix of $\boldsymbol{\Sigma}$. Let $\mathbf{z} = (\mathbf{v}^{\mathrm{T}}, v_1\mathbf{v}^{\mathrm{T}}, v_2v_{11}, v_2v_{12}, \ldots, v_2v_{20})^{\mathrm{T}}$ and $\mathbf{z}_i$'s are generated from the distribution of $\mathbf{z}$.

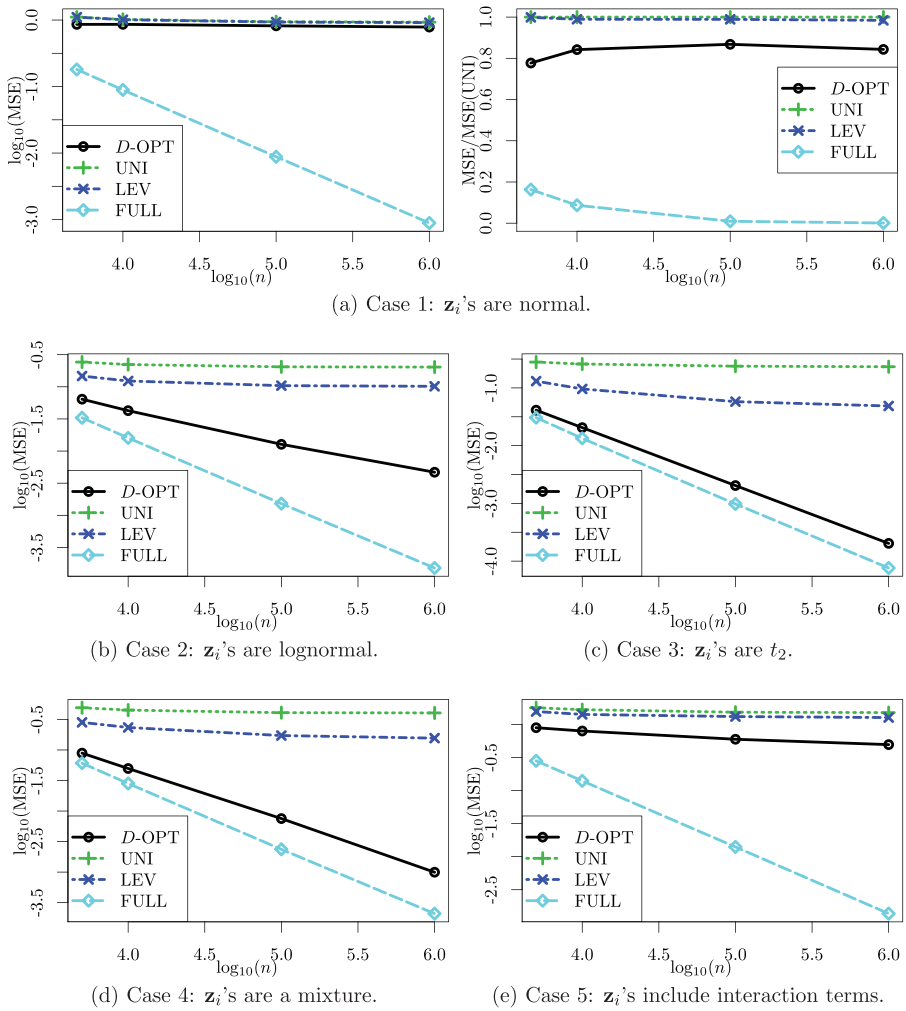
The simulation is repeated $S = 1000$ times and empirical mean squared errors (MSE) are calculated using $\mathrm{MSE}_{\beta_0} = S^{-1}\sum_{s=1}^{S}(\hat{\beta}_0^{(s)} - \beta_0)^2$ and $\mathrm{MSE}_{\boldsymbol{\beta}_1} = S^{-1}\sum_{s=1}^{S}\|\hat{\boldsymbol{\beta}}_1^{(s)} - \boldsymbol{\beta}_1\|^2$ for intercept and slope estimators from different approaches, where $\hat{\beta}_0^{(s)}$ and $\hat{\boldsymbol{\beta}}_1^{(s)}$ are estimates in the sth repetition. We compare four different approaches: D-OPT, the D-optimality-motivated IBOSS algorithm described in Algorithm 1 (black solid line (—○—)), UNI (green short dotted line (···+···)), LEV (blue dashed line (·-✕-·)), and FULL, the full data approach (aqua long dashed line (—◇—)). To get the best performance of the LEV method in parameter estimation, exact statistical leverage scores are used to calculate the subsampling probabilities. Note that for linear regression, the divide-and-conquer method produces results that are identical to these from the FULL (Lin and Xie 2011; Schifano et al. 2016), while the computational cost is not lower than the FULL. Thus, the comparisons between the IBOSS approach and the full data approach reflect the relative performance of the IBOSS and the divide-and-conquer method in the context of linear regression.

For full data sizes $n = 5 \times 10^3$, 10^4, 10^5, and 10^6 and fixed subdata size $k = 10^3$, Figures 1 and 2 present plots of the $\log_{10}$ of the MSEs against $\log_{10}(n)$. Figure 1 gives the $\log_{10}$ of the MSEs for estimating the slope parameter $\boldsymbol{\beta}_1$ using different methods. As seen in the plots, the D-OPT IBOSS method uniformly dominates the subsampling-based methods UNI and LEV, and its advantage is more significant if the tail of the covariate distribution is heavier. More importantly, the MSEs from the D-OPT IBOSS method for estimating $\boldsymbol{\beta}_1$ decrease as the full data sample size n increases, even though the subdata size is fixed at $k = 10^3$. For the normal covariate distribution in Figure 1(a), the decrease in the MSE for the D-OPT IBOSS estimator is not as evident because, as shown in Theorem 6, the convergence rate of variances for this case is as slow as $p/k/\log n$. To show the relative performance of the D-OPT IBOSS approach compared to that of the subsampling-based approaches, in the right panel of Figure 1(a), we scale all the MSEs so that the MSEs for the UNI method are one. From this figure, the MSEs for the D-OPT IBOSS approach are about 80% of those for the subsampling-based approaches.

Unlike the IBOSS method, the random subsampling-based methods yield MSEs that show very little change with increasing n except for the t_2 and mixture covariate distributions. This agrees with the conclusion in Theorem 1 that the covariance matrix for a random subsampling-based estimator is bounded from below by a matrix that depends only on the subdata size k if the fourth moment of the covariate distribution is finite. For the t_2 and mixture covariate distributions, the second moments of the covariate distributions are not finite and we see that the MSEs decrease as n becomes larger. However, the convergence rates are much slower than for the IBOSS method.

For the full data approach, where all data points are used, the MSEs decrease as the size n increases. It is noteworthy that the performance of the D-OPT IBOSS method can be comparable to that of using the full data for estimating $\boldsymbol{\beta}_1$. For example, as shown in Figure 1(d) for the mixture of distributions, an analysis using the D-OPT IBOSS method with subdata size $k = 10^3$ from full data of size $n = 10^6$ outperforms a full data analysis with data of size $n = 10^5$; the MSE from the full data analysis is 2.4 times as large as the MSE from using subdata of size $k = 10^3$.

Figure 2 gives results for estimating the intercept parameter β_0. In general, the D-OPT IBOSS method is superior to other subdata-based methods, but its MSE does not decrease as the full data size increases. This agrees with the result in Theorem 4. We also calculate the MSE of the adjusted estimator in (25), $\hat{\beta}_0^{Da} = \bar{y} - \bar{\mathbf{z}}^{\mathrm{T}}\hat{\boldsymbol{\beta}}_1^{\mathrm{D}}$, which is labeled with D-OPTa (red dashed



(a) Case 1: $\mathbf{z}_i$'s are normal.

(b) Case 2: $\mathbf{z}_i$'s are lognormal.

(c) Case 3: $\mathbf{z}_i$'s are t_2.

(d) Case 4: $\mathbf{z}_i$'s are a mixture.

(e) Case 5: $\mathbf{z}_i$'s include interaction terms.

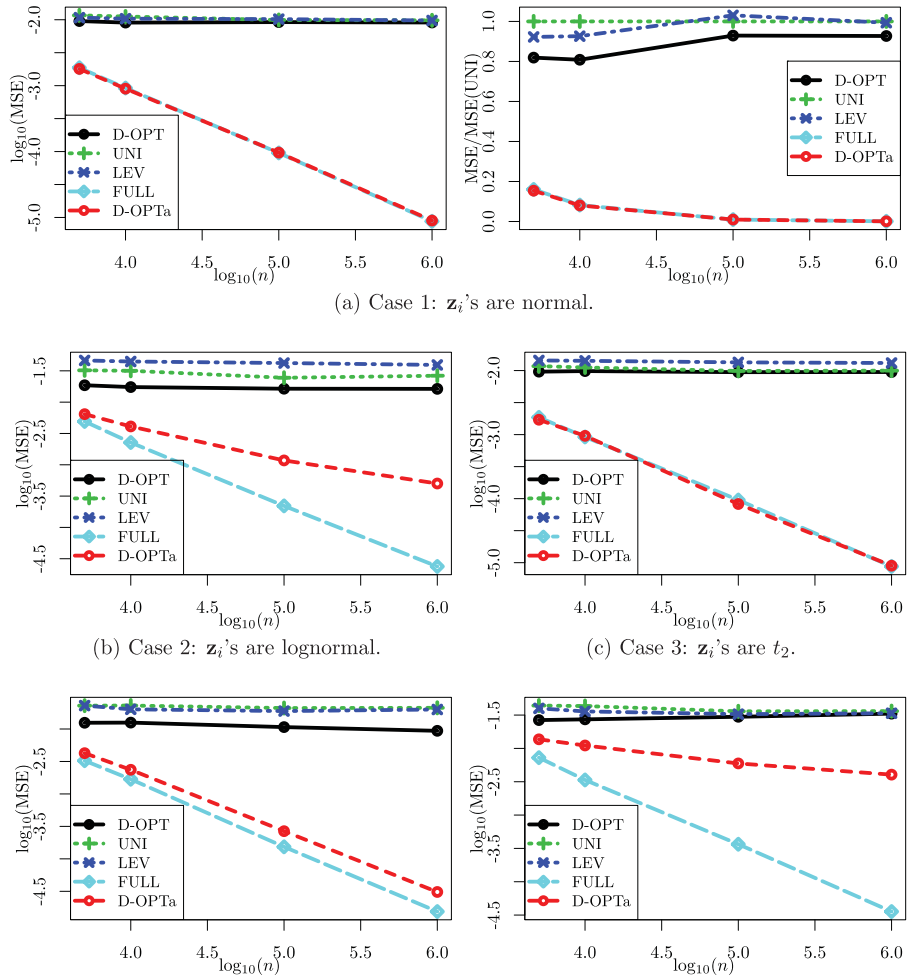
Figure 1. MSEs for estimating the slope parameter for five different distributions for the covariates $\mathbf{z}_i$. The subdata size k is fixed at $k = 1000$ and the full data size n changes. Logarithm with base 10 is taken of n and MSEs for better presentation of the figures except for the right panel of (a) in which MSEs are scaled so that MSEs for the UNI method are 1.

line (- ⊖ -)) in Figure 2. It is seen that the relative performances of $\hat{\beta}_0^{Da}$ for Cases 2, 4, and 5 are similar to those of the slope estimator, which agrees with the asymptotic properties discussed below (25) in Section 4. For Cases 1 and 3, the results are very interesting in that $\hat{\beta}_0^{Da}$ performs as good as the full data approach, which means that the convergence rate of $\hat{\beta}_0^{Da}$ is much faster than that of the slope estimator. This seems surprising, but it agrees with the asymptotic properties discussed below (25) in Section 4. For these two cases, $\mathrm{E}(\mathbf{z}) = \mathbf{0}$ and this is the reason why the convergence rate can be faster than that of the slope parameter estimators.

To see the effect of the subdata size k for estimating the slope parameter $\boldsymbol{\beta}_1$, Figure 3 presents plots of the $\log_{10}$ of the MSEs against the subdata size k, with choices $k = 200$, 400, 500, 10^3, 2×10^3, 3×10^3, and 5×10^3, for fixed full data size $n = 10^6$. The MSE for using the full data is a constant with respect to k and is plotted for comparison. Clearly, all subdata-based methods improve as the subdata size k increases, with the D-OPT IBOSS method again being the best performer. For example, when $n = 10^6$ for the mixture covariate distribution, the analysis based on the D-OPT IBOSS method with $k = 200$ is about 10 times as accurate as that of the LEV method with $k = 5 \times 10^3$ as measured by the MSE value.

To evaluate the performance of the D-OPT IBOSS approach for statistical inference, we calculate the empirical coverage probabilities and average lengths of the 95% confidence intervals from this method. Results for the full data analysis are also computed for comparison. Figure 4 gives results for the normal and mixture covariate distributions. The estimated parameter is the first slope parameter β_1. Confidence intervals are constructed using $\hat{\beta}_1^{(s)} \pm Z_{0.975}\mathrm{SE}_1^{(s)}$, where $\hat{\beta}_1^{(s)}$ and $\mathrm{SE}_1^{(s)}$ are the estimate and its standard error of β_1 in the sth repetition, and $Z_{0.975}$ is the 97.5th percentile of the standard normal distribution. It is seen that all empirical coverage levels are close to the nominal level of 0.95, which shows that the inference based on IBOSS subdata is valid. We do not compare this to subsampling-based approaches because we are not aware of theoretically justified methods for constructing confidence intervals under these approaches.

Results on computational efficiency of the D-OPT IBOSS approach are presented in Table 2, which shows CPU times (in seconds) for different combinations of the full data size n and the number of covariates p for a fixed subdata size of $k = 10^3$ and normal distribution for the $\mathbf{z}_i$'s. The R programming language (R Core Team 2015) is used to implement each method. For the IBOSS approach, it requires a partition-based partial sort



(a) Case 1: $\mathbf{z}_i$'s are normal.

(b) Case 2: $\mathbf{z}_i$'s are lognormal.

(c) Case 3: $\mathbf{z}_i$'s are t_2.

(d) Case 4: $\mathbf{z}_i$'s are a mixture.

(e) Case 5: $\mathbf{z}_i$'s include interaction terms.

Figure 2. MSEs for estimating the intercept parameter for five different distributions for the covariates $\mathbf{z}_i$. The subdata size k is fixed at $k = 1000$ and the full data size n changes. Logarithm with base 10 is taken of n and MSEs for better presentation of the figures except for the right panel of (a) in which MSEs are scaled so that MSEs for the UNI method are 1.

algorithm which is not available in R, so the standard C++ function *nthelement* (Stroustrup 1986), is called from R for partial sorting. To get good performance in terms of CPU times for the LEV method, the leverage scores are approximated using the fast algorithm in Drineas et al. (2012). The CPU times for using the full data are also presented for comparison. All computations are carried out on a desktop running Windows 10 with an Intel I7 processor and 16GB memory.

It is seen from Table 2 that the D-OPT IBOSS method compares favorably to the LEV method, both being more efficient than the full data method. Results for other cases are similar and thus are omitted.

Table 2. CPU times for different combinations of n and p with a fixed $k = 10^3$.

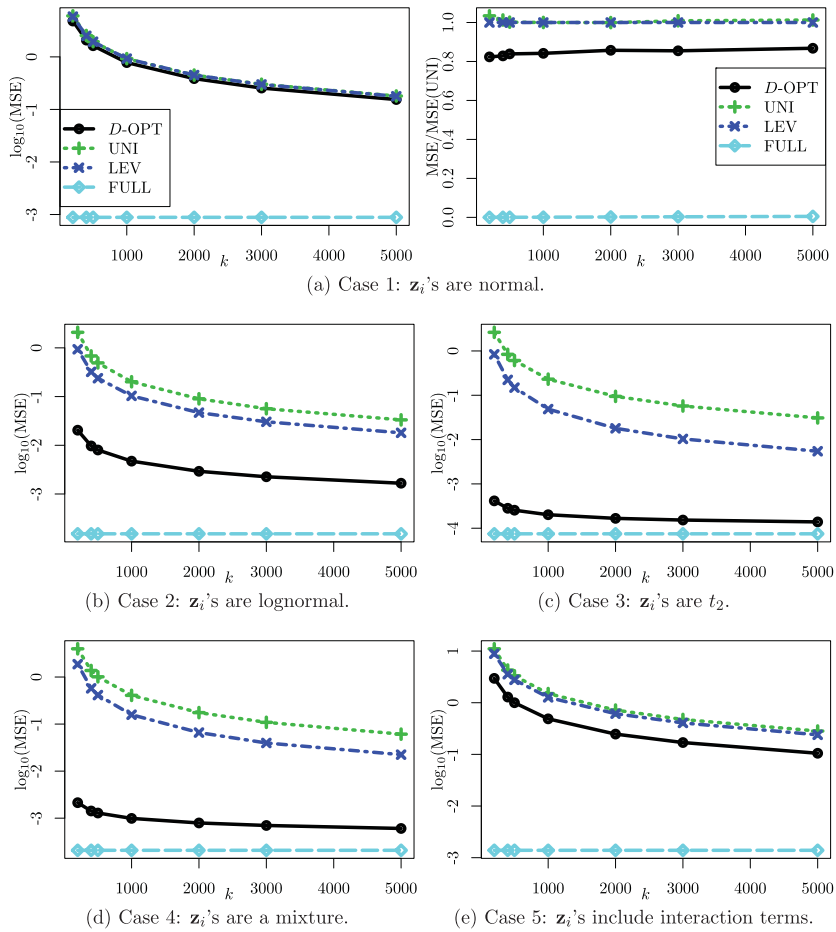
(a) CPU times for different n with $p = 500$				
n	D-OPT	UNI	LEV	FULL
5×10^3	1.19	0.33	0.88	1.44
5×10^4	1.36	0.29	2.20	13.39
5×10^5	8.89	0.31	21.23	132.04
(b) CPU times for different p with $n = 5 \times 10^5$				
p	D-OPT	UNI	LEV	FULL
10	0.19	0.00	1.94	0.21
100	1.74	0.02	4.66	6.55
500	9.30	0.31	21.94	132.47

5.2. Real Data

In this section, we evaluate the performance of the proposed IBOSS approach on two real data examples.

5.2.1. Example 1: Food Intakes Data

The first example is a dataset obtained from the *Continuing Survey of Food Intakes by Individuals* (CSFII) that was published by the Human Nutrition Research Center, U.S. Department of Agriculture, Beltsville, Maryland (CSFII Reports No. 85-4 and No. 86-3). Part of the dataset has been used in (Thompson et al. 1992). It contains dietary intake and related information for $n = 1,827$ individuals, such as the intakes of calorie, fat, protein, and carbohydrate, as well as body mass index, age, etc. The size of this dataset is not too big, and we can compare the IBOSS method to the full analysis. With this size of the data, we are also able to plot the full data to compare its pattern with that of the subdata selected by the IBOSS method. Interest is in examining the effects of the average intake levels of fat (z_1), protein (z_2), carbohydrate (carb, z_3), as well as body mass index (BMI, z_4) and



(a) Case 1: $\mathbf{z}_i$'s are normal.

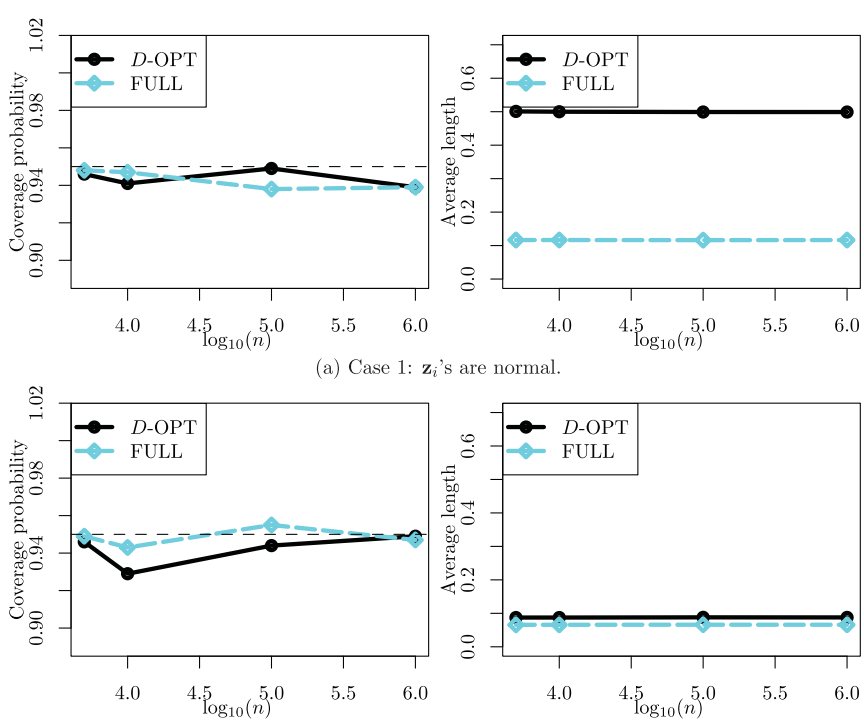
(b) Case 2: $\mathbf{z}_i$'s are lognormal.

(c) Case 3: $\mathbf{z}_i$'s are t_2.

(d) Case 4: $\mathbf{z}_i$'s are a mixture.

(e) Case 5: $\mathbf{z}_i$'s include interaction terms.

Figure 3. MSEs for estimating the slope parameter for five different distributions for the covariates $\mathbf{z}_i$. The full data size is fixed at $n = 10^6$ and the subdata size k changes. Logarithm with base 10 is taken of MSEs for better presentation of the figures except for the right panel of (a) in which MSEs are scaled so that MSEs for the UNI method are 1.



(a) Case 1: $\mathbf{z}_i$'s are normal.

(b) Case 4: $\mathbf{z}_i$'s are a mixture.

Figure 4. Empirical coverage probabilities and average lengths of 95% confidence intervals from the D-OPT IBOSS method and the full data method. The gray horizontal dashed line in the left panel is the intended coverage probability 0.95. The subdata size is fixed at $k = 10^3$.

Table 3. Estimation results for the CSFII data. For the D-OPT IBOSS method, the subdata size is $k = 10p = 50$.

	D-OPT		FULL	
Parameter	Estimate	Std. Error	Estimate	Std. Error
Intercept	33.545	46.833	45.489	11.883
Age	− 0.496	1.015	− 0.200	0.234
BMI	− 0.153	0.343	− 0.521	0.224
Fat	8.459	0.405	9.302	0.115
Protein	5.080	0.386	4.254	0.127
Carb	3.761	0.106	3.710	0.035

age (z_5) on calorie intake, y. Thus $p = 5$. We fit the model

$$y = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \beta_3 z_3 + \beta_4 z_4 + \beta_5 z_5 + \varepsilon,$$

using both the D-OPT IBOSS method with $k = 10p = 50$ and the full data. Results are summarized in Table 3. The D-OPT IBOSS estimates for the slope parameters are not very different from those from the full data, and the signs of the estimates from the IBOSS method and from the full data are consistent. The standard errors for the IBOSS method, while larger than for the full data, are reasonably good in view of the small subdata size. The estimates for the intercept parameter show a larger difference, and the standard error for the IBOSS method is large. This agrees with the theoretical result that the intercept cannot be estimated precisely without a large subdata size. The D-OPT IBOSS method identifies the significant effects of fat, protein, and carbohydrate intake levels on calorie intake. Based on the full data, the effect of BMI is near the boundary of significance at the 5% level, and is not identified as significant by the D-OPT IBOSS method.

Figure 5 gives scatterplots of calorie intake against each covariate for the full data of $n = 1827$ with the D-OPT subdata of $k = 50$ marked. It is seen that the relationship between the response and each covariate is similar for the subdata and the full data, especially for covariates fat, protein, and carbohydrate. Also, there do not seem to be any extreme outliers in this dataset.

To compare the IBOSS performance to that of the subsampling approaches, we compute the MSE for the vector of slope parameters for each method by using 1000 bootstrap samples. Each bootstrap sample is a random sample of size n from the full data using uniform sampling with replacement. For a bootstrap sample, we implement each subdata method to obtain the subdata estimate or implement the full data approach to obtain the full data estimate. The bootstrap MSEs are the empirical MSEs

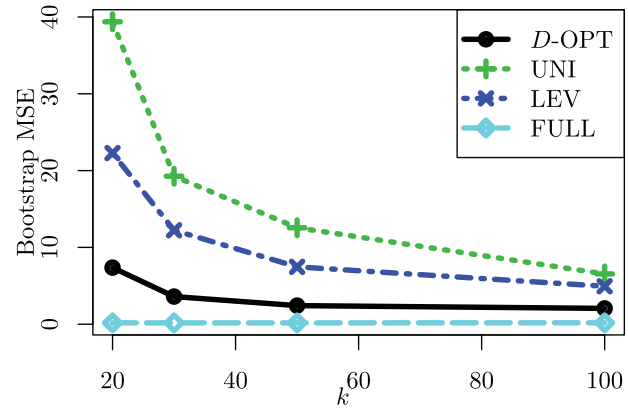


Figure 6. MSEs for estimating slope parameters for the CSFII data. They are computed from 1000 bootstrap samples.

corresponding to the 1000 estimates. We do this for $k = 4p$, $6p$, $10p$, and $20p$. Figure 6 shows that the D-OPT IBOSS method dominates random subsampling-based methods. The full data approach is shown for comparison.

5.2.2. Example 2: Chemical Sensors Data

In this example, we consider chemical sensors data collected to develop and test strategies to solve a wide variety of tasks, for example, to develop algorithms for continuously monitoring or improving response time of sensory systems (Fonollosa et al. 2015). The data were collected at the ChemoSignals Laboratory in the BioCircuits Institute, University of California San Diego. It contains the readings of 16 chemical sensors exposed to the mixture of Ethylene and CO at varying concentrations in air. Each measurement was constructed by the continuous acquisition of the sixteen-sensor array signals for a duration of about 12 hr without interruption. The concentration transitions were set at random times and to random concentration levels. Further information about the dataset can be found in Fonollosa et al. (2015).

For illustration, we use the reading from the last sensor as the response and readings from other sensors as covariates. Since trace concentrations often have a lognormal distribution (Goodson 2011), we take a log-transformation of the sensors readings. Readings from the second sensor are not used in the analysis because about 20% of the values are negative for reasons unknown to us. Thus, there are $p = 14$ covariates in this example. In addition, we exclude the first 20,000 data points

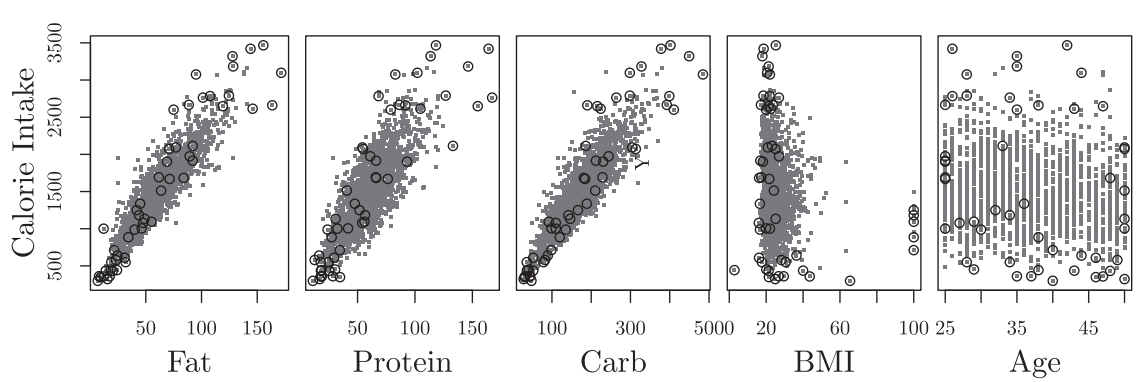


Figure 5. Scatterplots of calorie intake against each covariate for the full CSFII data (gray dots). The D-OPT subdata is labeled by ○.

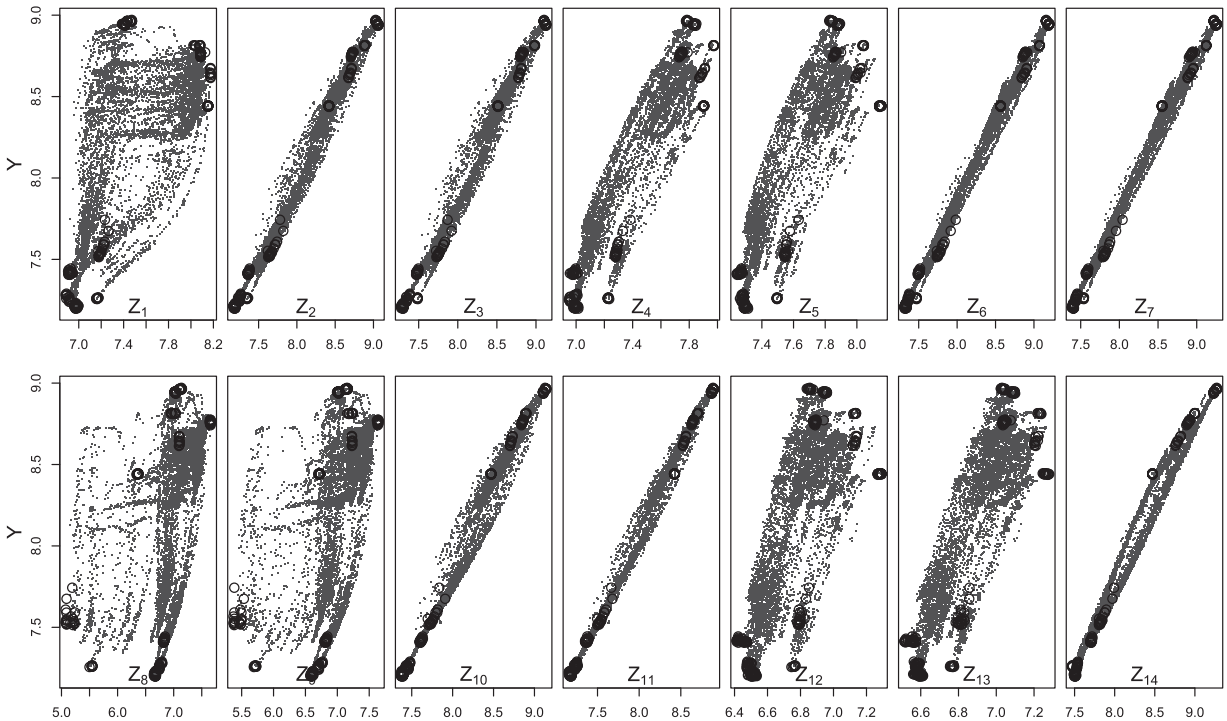


Figure 7. Scatterplots of a size 10,000 simple random sample of the chemical sensors data (gray dots). The D-OPT subdata of size $k = 280$ is plotted as ○.

corresponding to less than 4 min of system run-in time. Thus, the full data used contain $n = 4,188,261$ data points.

Figure 7 gives scatterplots of the response variable against each covariate for a simple random sample of size 10,000, with D-OPT subdata of $k = 280$ overlaid. Due to the size of the data, we cannot plot the full data in Figure 7. However, a simple random sample with a large sample size should be able to represent the overall pattern of the full data. It is seen that a linear model seems appropriate for the log-transformed readings and the relationship between the response and each covariate is similar for the subdata and the full data.

We also use bootstrap to calculate the MSEs of different estimators for estimating the slope parameters. As for the first example, we considered $k = 4p$, $6p$, $10p$, and $20p$ as subdata size for each method. Results computed from 100 bootstrap samples are plotted in Figure 8. The performance of the D-OPT IBOSS method lies between the full data approach and the subsampling-based methods.

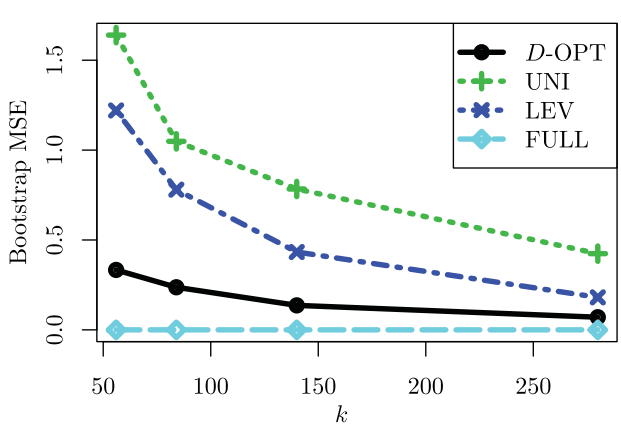


Figure 8. MSEs from 100 bootstrap samples for estimating slope parameters for the chemical sensors data.

6. Concluding Remarks

In this article, we have developed a subdata selection method, IBOSS, in the context of big data linear regression problems. Using the framework for the IBOSS method, we have analyzed existing subsampling-based methods and derived a lower bound for covariance matrices of the resultant estimators. For the IBOSS method, we focused on D-optimality. After a theoretical characterization of the IBOSS subdata under D-optimality, we developed a computationally efficient algorithm to approximate the optimal subdata. Theoretical properties of the D-OPT IBOSS method have been examined in detail through asymptotic analysis, and its performance has been demonstrated by using simulated and real data.

There are important and unsolved questions that require future study. For example, while we only considered the D-optimality criterion, there are other optimality criteria with meaningful statistical interpretations and different inferential purposes. This includes A-optimality, which seeks to minimize the average variance of estimators of regression coefficients, and c-optimality, which minimizes the variance of the best estimator of a prespecified function of the model parameters. These optimality criteria may also be useful to develop efficient IBOSS methods.

Identifying informative subdata is important for extracting useful information from big data and more research is needed. We hope that this work will stimulate additional research in the direction suggested in this article.

Supplementary Materials

The supplementary materials include an appendix with technical details and a document with additional numerical evidence for the performance of the IBOSS method.

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